2. The carriage should be moved longitudinally obtained feed through the leadscrew of the lathe.
3. A definite ratio between the longitudinal feed and rotation of the head stock spindle should be found out. Suitable gears with required number of teeth should be mounted on the spindle and the leadscrew.

Calculation for change – gear

$$\frac{\text{TPI on leadscrew}}{\text{TPI on work}} = \frac{\text{driver teeth}}{\text{driven teeth}}$$

By using the above formula, calculate the thread to be cut and connect the gear on the spindle and the gear on leadscrew.

4. A proper thread cutting tool is selected according to the shape of the thread. It is mounted on the toolpost with its cutting edge at the lathe axis and perpendicular to the axis of the work.
5. The position of the tumbler gears are adjusted according to the type of the thread (right hand or left hand).
6. Suitable spindle speed is selected and it is obtained through back gears.
7. Half nut lever is engaged at the right point as indicated by the thread chasing dial.
8. Depth of cut is set suitably to allow the tool to make a light cut on the work.
9. When the cut is made for required length, the half nut lever is disengaged. The carriage is brought back to its original position and the above procedure is repeated until the required depth of the thread is achieved.
10. After the process of thread cutting is over, the thread is checked by suitable gauges.

ACTIVITIES

1. To make an arrangement the students to visit Government Industrial Training Institute, Polytechnics and Engineering colleges nearby the school and learn about the types of lathe and its working principle.
2. To give practice to the students performed different types of operations in a lathe.

Questions

Part – I

Choose the correct answer

1. Lathe was invented by
   a. Henry Maudslay
   b. Eli Whitney
   c. James Nasmith
   d. Michael Faraday
2. The tool used in lathe is a
   a. Multipoint cutting tool
   b. Single point cutting tool
   c. Slitting saw tooth cutting tool
   d. Parting tool

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